

SOLUTION ARCHITECTURE

Date	10 March 2026
Team Project name	Streamlining Ticket Assignment For Efficient Support Operations
Team ID	SWTID-2026-1605
Maximum mark	4 Marks

SOLUTION ARCHITECTURE:

The **solution architecture** describes the overall structure of the system and how different components interact to provide efficient ticket management and assignment. The architecture is designed to ensure smooth communication between users, the application, and the database while enabling automatic ticket assignment and monitoring.

The system follows a **three-layer architecture** consisting of the **presentation layer, application layer, and data layer**.

1. Presentation Layer (User Interface)

The presentation layer is the **front-end part of the system** that users interact with. It provides a user-friendly interface for customers, support agents, and administrators.

Functions of this layer include:

- Allow customers to **create support tickets**.
- Allow customers to **check ticket status**.
- Allow support agents to **view assigned tickets**.
- Allow administrators to **manage users and monitor tickets**.

Technologies used in this layer include **HTML, CSS, and JavaScript**, which help create interactive and responsive web pages.

2. Application Layer (Business Logic)

The application layer contains the **core functionality and business logic** of the system. It processes user requests and performs operations such as ticket assignment, validation, and communication with the database.

Key functions include:

- Processing ticket submissions.
- Automatically assigning tickets to available support agents.
- Managing ticket priorities and status updates.
- Sending notifications and updates to users.

This layer acts as the **bridge between the user interface and the database**. It ensures that the system operates efficiently and that data flows correctly between components.

3. Data Layer (Database)

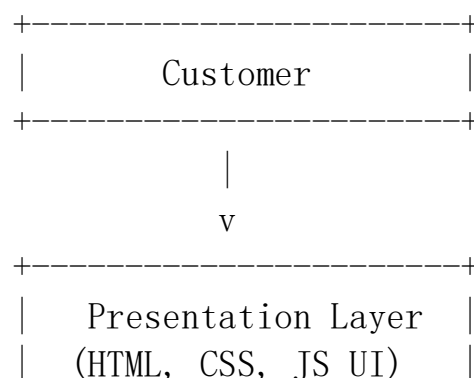
The data layer is responsible for **storing and managing system data**. All information related to tickets, users, and support agents is stored in the database.

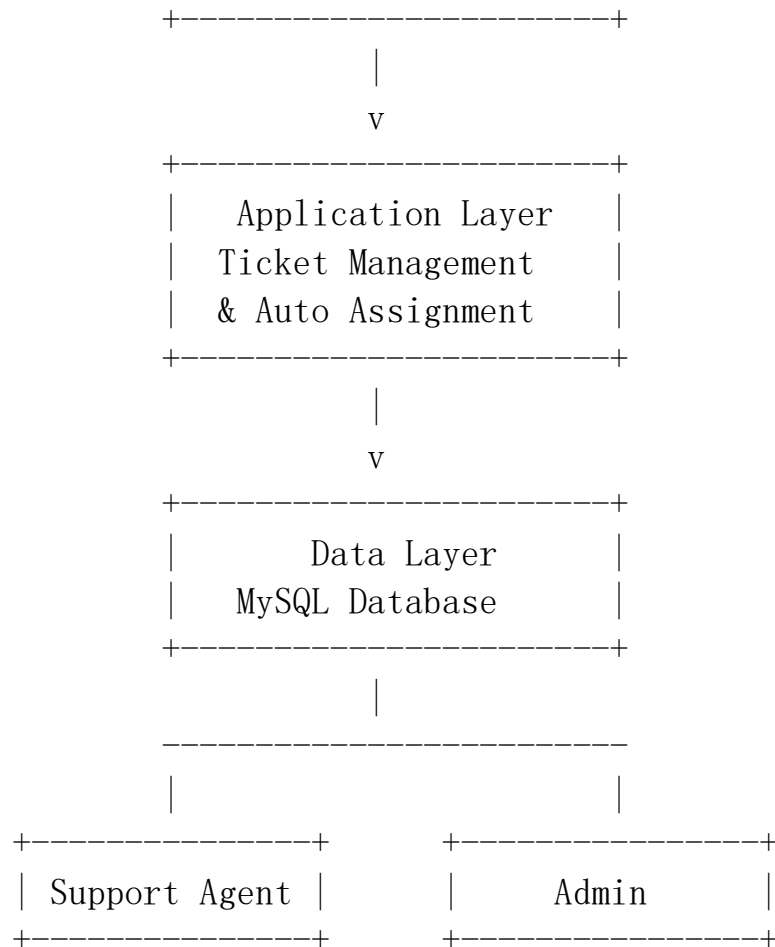
The database contains:

- Ticket details
- Customer information
- Support agent profiles
- Ticket status and history

A relational database such as **MySQL** is used to store and retrieve data securely.

Solution Architecture Diagram





Workflow of the Architecture

- The **customer submits a support ticket** through the user interface.
- The request is processed by the **application layer**, which validates the ticket information.
- The system checks the **database for available support agents** and their skills.
- The system automatically **assigns the ticket to the most suitable agent**.
- The **support agent resolves the issue and updates the ticket status**.
- The **customer receives updates and the final solution** through the system.